



Cambridge International AS & A Level

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MATHEMATICS

9709/15

Paper 1 Pure Mathematics 1

October/November 2025

1 hour 50 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.





- (a)** State the equation of the circle.

[2]

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- (b) The circle passes through the point $(8, k)$.

Find the two possible values of k .

[3]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



2 A geometric progression has first term $3 + 4\sqrt{2}$ and second term $5 - \sqrt{2}$.

- (a) Find the common ratio of the geometric progression. Give your answer in the form $\sqrt{2} + p$, where p is an integer to be found. [3]

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- (b) Find the sum to infinity of the geometric progression. [2]

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- 3 (a) Express $4x^2 + 10x + 6$ in the form $a(x+b)^2 + c$, where a , b and c are rational constants to be determined. [2]

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- (b) The curve with equation $y = 4x^2 + 10x + 6$ and the line $y = k$ have exactly one point of intersection.

Using your answer to part (a) or otherwise, state the value of the constant k . [1]

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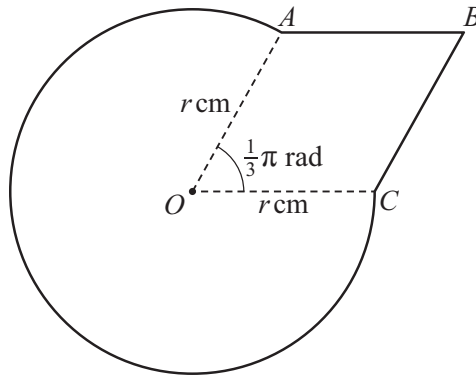
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The diagram shows the design for a company's new logo. The sector of the circle, centre O , has radius r cm. The acute angle $AOC = \frac{1}{3}\pi$ radians. The quadrilateral $OABC$ is a rhombus.

- (a) Find an expression for the perimeter of the design. Give your answer in terms of π and r . [2]

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- (b)** It is now given that the perimeter of the design is 200 cm.

Find the area of the design. Give your answer to 3 significant figures. [5]

[illegible]



5 Solve the equation

$$x^3 - 28 + \frac{27}{x^3} = 0.$$

[3]

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6 (a) Show that the equation

$$6 \sin \theta + \frac{1}{\tan \theta} = \frac{4}{\sin \theta}$$

can be written in the form

$$6 \cos^2 \theta - \cos \theta - 2 = 0. \quad [3]$$

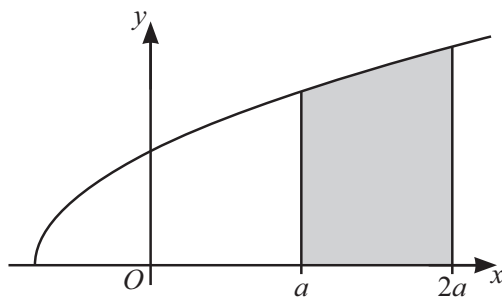
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[3]

[illegible]

[2]

[illegible]



The diagram shows the curve with equation $y = \sqrt{6x+5}$. The shaded region is bounded by the curve, the x -axis and the lines $x = a$ and $x = 2a$, where a is a positive constant.

The shaded region is rotated through 360° about the x -axis to form a solid. The solid has volume, V , such that $V \geq 46\pi$.

- (a) Show that $9a^2 + 5a - 46 \geq 0$. [4]

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- (b) Find the range of possible values of a . [3]

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- 9 In the expansion of $(p + qx)^4$, the coefficient of x is equal to the coefficient of x^2 . The constants p and q are both positive.

(a) Find the ratio $p : q$. Give your answer in its simplest form. [3]

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- (b) It is given that the coefficient of x^3 is 486.

Find the values of p and q . [4]

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10 The function f is defined by

$$f(x) = 3 + \frac{7}{x-2}$$

for $x > 2$.

(a) It is given that $f(a) = 4$.

Find the value of a . [2]

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(b) Find an expression for $f^{-1}(x)$ and state the domain of f^{-1} . [4]

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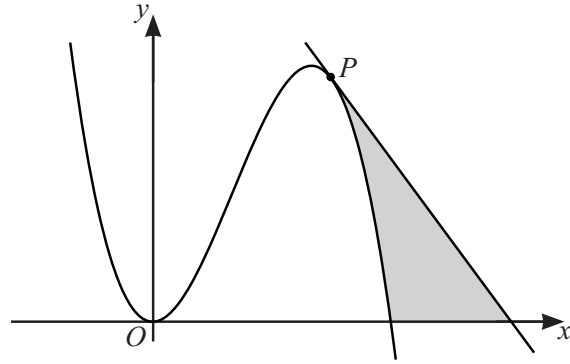
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The diagram shows the curve with equation $y = 4x^2 - x^3$ and the tangent to the curve at the point P . The point P has x -coordinate 3.

- (a)** Find the equation of the tangent to the curve at the point P . Give your answer in the form $y = mx + c$. [5]

[illegible]



- (b)** The shaded region is bounded by the curve, the x-axis and the tangent to the curve at P .

Find the exact area of the shaded region.

[6]

[illegible]

The graph of $y = 4x^2 - x^3$ is transformed by a stretch of scale factor $\frac{1}{3}$ in the x -direction. The point Q is the image of P under this transformation. The transformed shaded region is bounded by the transformed curve, the x -axis and the tangent to the transformed curve at Q .

- (c) (i)** Find the equation of the transformed curve in the form $y = mx^2 + nx^3$, where m and n are integers to be found. [1]

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- (c) (ii) State the coordinates of Q and the area of the transformed shaded region. [2]

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Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, creating a series of uniform gaps for letter height. The lines are evenly spaced across the entire page, providing a guide for consistent letter formation. There is no text or other markings on the page.

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